

Distributions

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Your focus should be on **foundations and intuition**, not formal proofs.

1. What Is a Distribution?

- A generalized function—something that behaves like a function but isn't one in the traditional sense
- Used when usual functions aren't flexible enough (e.g., derivatives of non-smooth functions)
- A linear functional which is understood only by its EVALUATION on a test function.

2. Distributions as Linear Functionals

Distributions act as linear functionals on test functions. This means that it linearly maps test functions to the space of real numbers.

- Test functions are both smooth and compactly supported. Smooth means that they have infinitely many derivatives. Compactly supported means that they evaluate to zero outside of a compact set.
- Distributions are linear. For distribution T , test functions f and g , and scalars a and b :

$$T[af + bg] = aT[f] + bT[g]$$

- Distributions aren't evaluated at test functions, they **act** on test functions. To perform this action, we integrate the test function with a function associated with the distribution.

$$T_\phi[f] = \int_{-\infty}^{\infty} \phi(x)f(x)dx$$

3. The Dirac delta “function”

The Dirac delta function is not a function, really, but a distribution, due to it's discontinuous nature.

- Defined informally

We can informally think of a Dirac delta function like a Gaussian with a width that in the limit approaches zero... but this analogy only holds in the context of integrating over the delta function.

- Written out, it is:

$$\int_{-\infty}^{\infty} f(x)\delta(x-a)dx = f(a)$$

Which is to say that unless we are at the location where $\delta(0)$ is valid, the delta “function” equals zero.

Note that $\delta(x-a)$ is only nonzero at a . If, when handling a bounded integral, a is outside of the bounds, then the delta function is never satisfied and hence the integral is zero.

4. How Are Distributions Used? Distributions are more general than functions. There are many scenarios in PDEs for example where a general solution to a PDE is not important and might not exist. In these scenarios we can skirt past this problem by looking for weak solutions, using test functions, and other clever tricks. Although these will produce non-general solutions, we often do not care. In physics, for instance, we typically have very elegant and drastically simplifying scenarios that allow us to freely pretend that our weak solutions are actually just solutions.

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- Recall Integration by Parts:

$$\int_a^b h(x)f'(x)dx = h(x)f(x)\Big|_a^b - \int_a^b h'(x)f(x)dx$$

We can think of weak derivatives like *anti* integrals, meaning we assume to know:

$$\int_a^b h'(x)f(x)dx + \int_a^b h(x)f'(x)dx$$

and therefore find:

$$\int_a^b h'(x)f(x)dx + \int_a^b h(x)f'(x)dx = h(x)f(x)\Big|_a^b.$$

The advantage here is that the criteria for integrability is much weaker than differentiability. Jump Discontinuities are thus weak-differentiable.

- Green's Functions:

Green's functions are really distributions. Linear differential operators (like $\frac{d}{dx}$) operate on them. These distributions then result in Dirac delta functions. The purpose of a Dirac delta function is to be integrated—thus, we express the greens functions as:

$$\int LG(f(x))dx = \int \delta(x)f(x) = f(0).$$

- Fourier transform in discrete and continuous spaces... just the equations, I think, and maybe a drawing.

Fourier transforms map a function in one space to another. In physics, we tend to consider this mapping as between position and momentum space. They are defined as $f \rightarrow g$, where:

$$g(k) = \int_{-\infty}^{\infty} e^{i(2\pi k)x} f(x)dx$$

or, in the discrete case:

$$g(k) = \sum_{n=0}^{N-1} x_n e^{-i(2\pi k)n/N}$$

5. A Simple Example

We will now define the weak derivative of the delta function. Observe that:

$$\delta[f'] = \int_a^b \delta(x)f'(x)dx = \delta(x)f(x)\Big|_a^b - \int_a^b \delta'(x)f(x)dx = 0 - \delta'[f]$$

This implies $\delta[f'] = -\delta'[f] \implies \delta'[f] = -\delta[f']$

Show how the derivative of the Heaviside function (step function) is the Dirac delta. We will do this using integration by parts. We will define the distributional derivative of a distribution, $H(f) = \int_a^b h(x)f'(x)dx$, as the distribution, $G(f) = \int_a^b g(x)f(x)dx$, which satisfies the relationship:

$$\int_a^b h(x)f'(x)dx = h(x)f(x)\Big|_a^b - \int_a^b g(x)f(x)dx$$

We will now show that this relationship is fulfilled by the Heaviside step function, $H(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}$, whose distributional derivative is the delta function, $\int_{-\infty}^{\infty} \delta(x)f(x)dx = f(0)$.

Observe that:

$$\int_a^b H(x)f'(x)dx = \begin{cases} 0 & a < b < 0 \\ f(b) - f(0) & a < 0 < b \\ f(b) - f(a) & 0 < a < b \end{cases}$$

Observe that delta can be used in the right hand side of integration by parts to yield the same answer:

$$H(x)f(x)\Big|_a^b - \int_a^b \delta(x)f(x)dx = \begin{cases} 0 \cdot f(b) - 0 \cdot f(a) - 0 = 0 & a < b < 0 \\ 1 \cdot f(b) - 0 \cdot f(a) - f(0) = f(b) - f(0) & a < 0 < b \\ 1 \cdot f(b) - 1 \cdot f(a) - 0 = f(b) - f(a) & 0 < a < b \end{cases}$$

As these two piecewise functions are identical the integration by parts relationship between the Heaviside distribution and the delta distribution holds. This means that is reasonable to claim that the delta function is a weak derivative of the Heaviside step function.

Another proof of this relationship can be seen here. Observe that:

$$\int_{-\infty}^x \delta(x)f(x)dx = \begin{cases} 0 & x < 0 \\ f(0) & x > 0 \end{cases} = H(x)f(0)$$

This implies that the Heaviside function is the weak antiderivative of the delta function.
